

EXP.NO: 6

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based Personal Portfolio Builder application by implementing a simple login system where the user's session is maintained.

ALGORITHM:

1. Input:

- The user enters username and password in the login form.
- The servlet checks the credentials and creates a session if valid.

2. Steps for Implementing Session Tracking using HttpSession:

1. Create the Login Form (HTML):

- Design an HTML page with input fields for username and password.
- Add a submit button to send data to the Login Servlet using POST method.

2. Create the Login Servlet:

- Develop a servlet to handle login requests.
- Retrieve username and password from the request.
- Validate the credentials (hardcoded values for simplicity).
- If valid, create a new session using HttpSession.
- Store the username in the session using setAttribute().

3. Validate User Credentials:

- Compare entered username and password with predefined values.
- If valid → create session and redirect to portfolio page.
- If invalid → display error message and reload login page.

4. Create and Track User Session:

- Use HttpSession object to maintain user session.
- Store user data like username in session.
- Session helps to track the user across pages.

5. Redirect to Portfolio Page:

- After successful login, redirect the user to the Portfolio Dashboard page.
- Check session in the dashboard page before displaying content.
- If session is not available, redirect back to login page.

6. Logout and Session Invalidation:

- Provide a logout option.
- When user clicks logout, invalidate session using session.invalidate().
- Redirect user back to login page.

7. Session Expiration:

- If session expires or user logs out, session data is removed.
- User must login again to access the portfolio page.

Program

Login.html

HTML

```
<!DOCTYPE html> <html> <head> <title>Smart City Portal Login</title> <style> body {
font-family: Arial; background: linear-gradient(to right, #667eea, #764ba2); display: flex;
justify-content: center; align-items: center; height: 100vh; } .login-box { background: white;
padding: 30px; border-radius: 10px; box-shadow: 0 0 15px rgba(0,0,0,0.2); width: 300px;
text-align: center; } input { width: 90%; padding: 10px; margin: 10px 0; border-radius: 5px;
border: 1px solid #ccc; } button { background: #667eea; color: white; border: none; padding:
10px; width: 100%; border-radius: 5px; cursor: pointer; } button:hover { background: #5a67d8;
} </style> </head> <body> <div class="login-box"> <h2>Smart City Portal Login</h2> <form
action="LoginServlet" method="post"> <input type="text" name="username" placeholder="Enter
Username" required> <input type="password" name="password" placeholder="Enter Password"
required> <button type="submit">Login</button> </form> </div> </body> </html>
```

LoginServlet.java

Java

```
import java.io.*; import javax.servlet.*; import javax.servlet.annotation.WebServlet; import
javax.servlet.http.*; @WebServlet("/LoginServlet") public class LoginServlet extends
HttpServlet { protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException { String username = request.getParameter("username");
String password = request.getParameter("password"); response.setContentType("text/html"); //
Simple authentication (hardcoded) if (username.equals("admin") && password.equals("1234")) {
HttpSession session = request.getSession(); session.setAttribute("user", username);
response.sendRedirect("SmartCityServlet"); } else { PrintWriter out = response.getWriter();
out.println("<h3>Invalid Username or Password</h3>"); out.println("<a href='Login.html'>Try
Again</a>"); } } }
```

SmartCityServlet.java (Your Dashboard Page)

Java

```
import java.io.*; import javax.servlet.*; import javax.servlet.annotation.WebServlet; import
javax.servlet.http.*; @WebServlet("/SmartCityServlet") public class SmartCityServlet extends
HttpServlet { protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException { response.setContentType("text/html"); PrintWriter out =
response.getWriter(); HttpSession session = request.getSession(false); if (session != null &&
session.getAttribute("user") != null) { String user = (String) session.getAttribute("user");
out.println("<html><head><style>"); out.println("body { font-family: Arial;
background:#f4f6f9; padding:20px; }"); out.println(".header { background:#333; color:white;
padding:15px; border-radius:8px; }"); out.println(".box { background:white; margin:15px 0;
padding:15px; border-radius:8px; box-shadow:0 0 10px rgba(0,0,0,0.1);} ");
out.println(".logout { display:inline-block; margin-top:20px; padding:10px; background:red;
color:white; text-decoration:none; border-radius:5px; }");
out.println("</style></head><body>"); out.println("<div class='header'><h2>Smart City
Portal</h2></div>"); out.println("<h3>Welcome " + user + "</h3>"); out.println("<div
class='box'>Report civic issues like potholes, water leaks and street light faults.</div>");
out.println("<div class='box'>Track real-time public transport and city bus
schedules.</div>"); out.println("<div class='box'>Pay utility bills and access government
services online.</div>"); out.println("<a class='logout' href='LogoutServlet'>Logout</a>");
out.println("</body></html>"); } else { response.sendRedirect("Login.html"); } } }
```

LogoutServlet.java

Java

```
import java.io.*; import javax.servlet.*; import javax.servlet.annotation.WebServlet; import
javax.servlet.http.*; @WebServlet("/LogoutServlet") public class LogoutServlet extends
HttpServlet { protected void doGet(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException { HttpSession session = request.getSession(false); if
(session != null) { session.invalidate(); } response.sendRedirect("Login.html"); } }
```

Output:

Result:

Session tracking using HttpSession is successfully implemented in the Personal Portfolio Builder project. User login session is created, maintained, and destroyed using servlets.